



Analyzing the impact of digitized-education toward the future of education: A comparative study based on students' evaluation of teaching data

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ABSTRACT

This study investigates the key factors that impact the *digitalized-education*, and what those mean for the new and emerging educational models of learning. The research is based on state-of-the-art education and assessment models for learning such as Descriptive decision theory, Futures Literacy (FLF), and 8-Affordances framework that theoretically addresses both the need to use multidisciplinary components in identifying the several challenges and opportunities to the use of digital technologies in education (digitized-education) and improvement of the student's learning outcomes. Through a quasi-experimental study design and comparative analysis of the students' evaluation of teaching (SET) dataset ($n = 3178$) collected in a higher education setting; the study applies a multivariate analysis of covariance (MANCOVA) and multiple comparisons (post-hoc) tests to determine the impact or association that the evaluation period (between 2019–2021) and type of school or discipline have on the students' learning performance and evaluation. The results show that students' satisfaction and experiences with digitized-education have exponentially grown in recent times, and mainly varied across the years, and marginally by discipline or school. In addition, the study empirically sheds light on the importance of the key findings and results by considering their pedagogical, technological, economic infrastructure, and ethical implications for the future of education and sustainable educational practice.

1. Introduction

Digital technology has enabled innovative and ground-breaking methods for teaching and learning (Miao, Holmes, Huang, & Zhang, 2023; OECD, 2022a; Miller, 2018). The resultant TEL-based (technology-enhanced) education (Okoye, Rodriguez-Tort, Escamilla, & Hosseini, 2021; Sen and Leong, 2020; Bälter, 2021) has not only become an integral element of the present-day education (Okoye et al., 2021; Talandron-Felipe, 2020), but has also become a phenomenon allied to the global sustainable development goals (SDG), such as life-long learning and quality education (UNESCO, 2020a; OECD, 2021a). The innovative educational models are now at the heart of the curriculum design and teaching/learning process by many higher institutions (OECD, 2021a; Facer and Selwyn, 2021). Existing studies have looked into what pedagogical opportunities and lessons are taken from the past

decades of implementing the digital or educational technologies for learning (Facer and Selwyn, 2021; Sønderlund, Hughes, & Smith, 2019; UNESCO, 2022a). Even the most extensively implemented teaching methods and innovations can have mixed or varying outcomes (Facer and Selwyn, 2021; Facer, 2022). Therefore, pointing the educators to the various challenges or limitations to effective teaching and learning with technology has proven to be of utmost importance (Facer and Selwyn, 2021).

Recent studies have highlighted both the several advantages and challenges of using digital technologies in modeling or delivering education, and have mentioned the need for strategies to reduce the achievement gap and improve learning outcomes for students (Cabral Gouveia, Menezes, & Neves, 2023; Cevikbas, Greefrath, & Siller, 2023). Although the *digitalized-education* (which the authors refer to as the use of digital technologies in education or for learning purposes) has

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remained unchanged, the growing interest and emerging technologies have only become a part of a rather slow but definite educational evolution that has experienced different challenges and essential blocks, particularly following the impact or disruptions caused by the recent global pandemic in education (Facer and Selwyn, 2021; Mercader and Gairín, 2020).

Amongst the many challenges to effective education and/or adoption of the digitalized-education, includes the limited or inability of educators to leverage information (educational dataset) that are recorded and stored in the database of the many institutions to support the learning processes and management, or for decision-making strategies and curriculum design, etc (LALA, 2020; Petersoni, Dumontii, Lafuente, & Lawiii, 2018; World Bank, 2021a; Lennox, Reuge, & Benavides, 2021). The so-called challenges and areas of opportunity do not only point toward the need for the development of *innovative educational frameworks or models* that integrate digital technologies for learning. But also highlights the need for methods or ways of making use of the "educational-generated datasets" to inform and improve the teaching/learning strategies or practices for the stakeholders (teachers, students, educational society).

1.1. The rationale of the study

The use of "educational-generated data" to inform the best ways or practices for managing and improving the educational processes has lately been seen as one of the most pertinent ways to achieving the quality of education and outcomes (LALA, 2020; World Bank, 2021a; Pettersson, 2020; Mulla, Denks, & Kalantzis, 2019; UNESCO, 2022b; Okoye et al., 2022a; UIS, 2023).

This current study believes that there are opportunities in developing methods that can extract useful information or patterns from the existing datasets which are recorded and stored in the database of the many institutions. The educational-generated data can be analyzed and transliterated into actionable plans or goals for improving the outcome of the teaching/learning process. Therefore, enabling the quality of education and delivery (OECD, 2022a; UNESCO, 2022a, 2022a). The research questions of this study is based on the premise and inaugural discussions and benchmark practices raised by the educational organizations and policymakers (UIS, 2023) concerning the learning process and assessment, as a way of offering valuable insights into the quality and equity of education by harnessing or uncovering important aspects/didactical questions that can be addressed using the educational data, such as the one done in this study.

The research questions of this study are as follows:

1. What factors impact the effective adoption of digitalized-education?
2. How can we analyze the educational dataset (e.g., student evaluation of teaching - SET) to understand key factors that can impact the students' learning process under the new learning normal or circumstances, and does the key factors differ by evaluation period and type of discipline?

The main contributions of this study are:

1. It discusses the didactic and socio-technical factors that impact the wide adoption of the digitized-education and/or influence the students' learning performance and outcome under the new learning circumstances.
2. It defines a data-structure (data-driven) approach used to understand the implications of the digitalized-education.
3. It determines the main factors that impact the students' evaluation of teaching across the years (between 2019-2021) and school/discipline.
4. It empirically sheds light on the implications of the digitized-education, and the use of educational-generated data (e.g., SET) to

inform the future directions of education and toward a sustainable educational practice.

This paper has two main components to its contribution. First, it discusses the didactic and socio-technical factors that impact the effective and wide adoption of the digitized-education and its implication for the future of education: from the pedagogical, technological, economic infrastructure, and ethical perspectives (Sections 2 and 5.1). In addition, it conducts a comparative analysis of the students' evaluation of teaching data collected through a survey in a higher education institution setting (see Sections 3 and 4). These were done not only to get a conceptual overview of the key aspects or impact of the digitalized-education toward an effective education and delivery, but also as a way of identifying the strengths and weaknesses, threats and opportunities for the concerned stakeholders (educators, teachers and students, educational community, policymakers, etc) (Section 5).

2. Background information

2.1. Digitalized-education: Pedagogical perspective

Determining the pedagogical impact of the use of digital technologies in education (EdTech) has remained a debate amongst the scientific community and organizations concerned with education and its transformation (OECD, 2021a; World Bank, 2021a; UNESCO, 2021a; OECD, 2022b; Fraisl et al., 2020; CONECTA, 2021a). For example, educational organizations such as UNESCO (UNESCO, 2022b), World Bank (World Bank, 2021a), and The OECD (OECD, 2022b) have in the past, particularly in the build-up to the recent global pandemic, called for different initiatives or ideas to explore the possibilities of "digital pedagogies" as a sustainable approach to reach and connect the millions of learners across the world, and scale up transformative learning and assessment (Mulla et al., 2019; Niel, 2022). The UNESCO's Education Research and Foresight (UNESCO, 2022c) whose target aims to better understand, anticipate, and imagine the future of knowledge and learning for the service of humanity, has also emphasized the future of education and global policy research to include: catalyzing global debates on how education and learning need to be reimagined in the context of constant technological changes toward achieving the quality of education ecosystem. The OECD (OECD, 2022b) has also pointed out key paradoxes for the future of education by identifying the main educational themes that must be considered when schooling for tomorrow. These comprise different challenges and opportunities where further research is considered to be most valuable, to name but a few; global innovation and development of new goals and modernization of the education ecosystem, to exploring the potential of virtual learning in education (OECD, 2022b).

Whereas there are many dimensions to the use of digital technologies in education particularly in the above areas, this study draws the readers' attention to the cutting-edge methods or practices that are deemed appropriate for the effective adoption of the "digitized-education for the future of education". For instance, those that can idiosyncratically be used to support the global goal of achieving the quality of education in its widest point of view. Indeed, educational organizations such as The UNESCO and World Bank (World Bank, 2021a; Mulla et al., 2019; UNESCO, 2021a; World Bank, 2022) have also investigated how technologies and their global adoption can contribute to "active pedagogy".

Instructional-wise, existing studies have researched tools and methods that help teachers to produce effective teaching, e.g., by discussing the Technological, Pedagogical, and Content Knowledge (TPACK) framework (Koehler, Mishra, & Cain, 2017; Tondeur, Scherer, Siddiq, Baran, & Tondeur, 2020) that methodologically deals with constructive ways of using digital technologies for teaching or learning content (Mulla et al., 2019). TEL-based models such as the TPACK represent as one of the promising ways of gaining an active pedagogy or

digital technologies in education (Mulla et al., 2019). Amongst the many dimensions are; (i) it represents as a didactical asset for communication and assessment of tasks in the classrooms, (ii) method for generation and disaggregation of data for learning/evaluation of the effectiveness of different learning strategies and outcomes, and (iii) a new subject or discipline that empowers knowledge and skills needed to create solutions to challenges of the society and education, and reimagining of the future of education (Mulla et al., 2019; UNESCO, 2021a, 2021b).

2.2. Future of education: Technological aspect

While it is critical to mention that the “digitized-education” and the several methodological interventions need to be adapted to the different countries or institutional contexts, on the other hand, a lot of countries and institutions are already implementing different learning tools and platforms to support the learners (UNESCO, 2021a). The authors note that the use of technologies in achieving quality education and transformation requires that educators must also seize the opportunity to make the learning platforms more inclusive, effective, and resilient than it was before, e.g., pre-, during, and post - the recent global pandemic (World Bank, 2021b). There is evidence that the congruence of the “technology” and “pedagogical transformations” can accelerate human-centered learning and recovery for the different stakeholders (teachers, students, educational community) (Okoye et al., 2021; Niel, 2022). Indeed, digital technologies and yet AI-based learning methods can introduce agile ways of delivering education and achieve “large-scale educational practices” that in turn promote lifelong learning (UNESCO, 2021a, b). To name but a few, the World Bank (World Bank, 2021a) under its strategy in understanding learning poverty (which refers to limited access to learning technologies, and losses experienced by the students e.g. during the recent Covid-19 pandemic) has continued in its efforts in creation and mobilization of ways to re-imagine education and operationalize the use of digital technologies for learning purposes. They aim to achieve the set goals through the discovery, diffusion, and deployment of new and emerging technologies under five main actions; (i) ask why e.g learning loss (ii) design for scale (iii) empowering teachers e.g. digital literacy and skills (iv) engaging the educational ecosystem, and (v) data-driven educational models and approaches for learning and assessment (World Bank, 2021a; Niel, 2022; Engzell, Frey, & Verhagen, 2021).

Furthermore, inclusive use and access to digital technologies for educational purposes are critical to achieving the global sustainable development goals (SDGs) (UN, 2020) particularly SDG4 which promotes the quality of education. Therefore, educators and concerned stakeholders must commit to rethinking the pedagogies and cater to the effective adoption of TEL-based education. Indeed, with learning poverty growing globally from 53% to 63% (World Bank, 2021b), some of the advocated technologies and practices aimed to handle the startling “digital divide” (Laufer et al., 2021; O’Hagan, 2022), to name but a few, includes: The development of AI and blended learning options for the students, Use of LMS (learning management systems) as central platform for teaching, learning, communication, advice, and support, EdTech for supporting remote and accelerated learning, Remote Formative Assessment solutions, Read@Home learning packages (books and other materials) to massively engage students and their parents at home, Compendium of Lesson plans and tools to improve reading instructions, and Technology for Teaching (T4T) by identifying scalable low-and-high-tech solutions for teachers’ professional development, and more importantly equitable educational funding and F.A.I.R. frameworks (Fit for purpose, Accessible, Inclusive, and Relevant-and-Responsive) that can promote openness, inclusiveness/agility, and cater for unequal access and strengthening of good practices on infrastructures and return on educational investments (OECD, 2022a; Okoye et al., 2021; OECD, 2021a; World Bank, 2021a; UNESCO, 2021a; World Bank, 2021b; UNESCO, 2021b; Okoye, Nganji, Escamilla, Fung, & Hosseini, 2022a; OECD, 2022c).

Recent studies on the incremental rise of digital technologies over the past two decades (between 1994 to 2019) have shown that it has significantly contributed to “opening up” education and scaled-up learning worldwide (Mulla et al., 2019; Niel, 2022). Ranging from the availability of e-contents, books, and materials by companies such as Amazon and Google in 1994 and 1998, to the birth of Wikipedia, Creative Commons, and MIT OpenCourseware (an open education movement) between 2001 and 2002. Those transitions were followed by Open Education Resources (OER) created at UNESCO’s Forum on the impact of Open Courseware for Higher Education for Developing Countries (UNESCO, 2022d) in the buildup to 2003. There was also the birth and scale of massively media distribution platforms for learning such as YouTube, Twitter, Facebook, and Instagram, between 2004 to 2010. AI-driven, personalized, and adaptive learning platforms/tools also led to the rise of Learning management systems (LMS) and Massively open online courses (MOOCs) such as edX, Coursera, Miriada, and FutureLearn between 2011 and 2013, and the so-called disruptive “second-order” innovation (augmented reality, virtual reality, simulations and holograms, flipped classrooms, m-learning, etc) (Groff, 2022). During those periods, the First UNESCO World Open Educational Resources (OER) Congress and Paris OER Declaration (UNESCO, 2022a) were formally adopted in 2012. Forward to the buildup of the recent global pandemic in 2020 (WHO, 2022), the Second World OER Congress and adoption of Ljubljana OER Action Plan (UNESCO, 2022b), enforcement of the 2016 European Union (EU) General Data Protection Regulation (2016/679) (European Union, 2022), and Beijing Consensus on AI and Education (UNESCO, 2022c) was among the many transformative and educational initiatives aimed at using the digital technologies in fostering education.

While the above-mentioned “technological” transitions reveal the paroxysm of the plethora of new tools and models of access to education. On the other hand, the key question has remained: Are educators and the concerned stakeholders pedagogically experiencing a movement toward an effective adoption of the digitalized-education? Perhaps, to ensure that the *innovations* that occurred during those periods are efficiently and effectively adopted, adapted, and implemented in the different educational settings and contexts; for example, the National Learning Platforms and Tools (UNESCO, 2021a) was a distance or online learning platform developed by the UNESCO’s Global Education Coalition (GEC) (UNESCO, 2021b) in its efforts to make available digital resources, learning tools, and materials for learners across the globe. Also noteworthy is the fact that UNESCO in those schemes aimed at tackling the educational disruptions and uncertainties that are anticipated to come with the post-pandemic education, are already rallying toward the future directions of education (UNESCO, 2021c), particularly as it concerns the strengths, weaknesses, opportunities, and threats that are associated with the digitalized-education (Facer and Selwyn, 2021; Mulla et al., 2019; UNESCO, 2022b; UN, 2020; UNESCO, 2022d). Interestingly, the OECD (OECD, 2022b) has mentioned that understanding the present (i.e., state-of-the-art in digital technologies and education done in this study) will help educators think about the future. Moreover, by reflecting on the “future of technology in education”, UNESCO has also set three key recommendations for educators with a focus on growing possibilities for openness and participation, reinforcement of control and reconnaissance (Mulla et al., 2019; UNESCO, 2021c; Montoya, McGreal, & Agbu, 2022) to include:

- Optimization of digital education, media, and resources affordance to improve learners’ engagement and outcomes.
- Development of digital textbooks, media, and resources for educational purposes that meet the needs of educators and learners, including their parents and home education.
- Ensure that educators continue to keep the different (digital) learning platforms/resources alive and flourishing for the benefit of students and the educational ecosystem at large.

2.3. Future of education: Economic and infrastructural aspect

In the bid toward recognizing education as the primary driver across all of the UN's 17 SDG goals, and the importance of promoting a holistic approach to education, the UN (UNESCO, 2022e) has not only conceded the need to adopt science, technology, and innovative (STI) strategies (UN, 2023) as a fundamental instrument for driving both national and global development through strengthening of collaboration and knowledge-sharing, scaling up ventures in STEM education, digital literacy and enhancement of technical, vocational, and tertiary education, distance education and training, and ensuring equal access to education and technology for all (UNESCO, 2022b; UN, 2022a, 2022a). But has also, on the other hand, recognized the importance of scaling up investments and international cooperation and partnerships (UN, 2022b), e.g., the Global Partnership for Education (GPE, 2022). Those action plans are staged to allow all stakeholders (educators, teachers, students, and the educational community) to gain a free, equitable, inclusive, and quality education that leads to effective learning outcomes or life-long learning. UNESCO under the Global Education Coalition (GEC) initiative (UNESCO, 2021b), which represents an alliance of over 200 public and private partners that are active in over 100 countries, encourages many institutions through leveraging hi-tech, low-tech, and even no-tech approaches, to mobilize, coordinate, match, and provide on-the-ground solutions (both local or global) for TEL-based education (Okoye et al., 2021; Khojasteh, Hosseini, & Nasiri, 2021; Rotar, 2022). They (UNESCO) have recently pronounced that never before has the educational system witnessed disruption at a large scale, and *partnership* amongst the stakeholders (government, policymakers, educators, financial investors) is perceived as the only way forward. The anticipated economic and infrastructural development can help to foster *digitalized-education*, e.g., digital learning solutions such as UNESCO's "Learning without Limits" project (Chekyravava, 2022) which have shown to support the teachers and make available learning materials and online socio-emotional aid to help the students and their teachers.

Furthermore, whereas *technology* represents \$350 billion (USD) of the global market, and is estimated to grow to more than \$3.2 trillion (USD) by 2025 (UN, 2023); the UN (UN, 2023) has confirmed as of the end of 2021, the average gap in income per capita between the developed and developing countries to be around \$40,749 (USD). They projected it cost the global economy around \$2 trillion (USD) during the recent global pandemic, and a slash of nearly \$8.5 trillion (USD) in output for the next two years (UN, 2021; Kaye et al., 2020). In Education, with over 500 million students estimated to have no access to remote learning or yet digitized-education (Facer and Selwyn, 2021; UN, 2022c), and learning poverty/loss increasing from 53% to 63% globally (World Bank, 2021b), the UN (UN, 2023) has highlighted the risk in lower-income countries falling behind in adoption of education 4.0 or industry 4.0. Thus, widening the already deepening economic, technology, and infrastructural gap or "digital divide" per se (Talan-dron-Felipe, 2020; Engzell et al., 2021; Laufer et al., 2021).

This study acknowledges the fact that notwithstanding the challenges posed by the various economies or disparities in the digitalized-education; several educational organizations and institutions are also working together, financially and in partnerships, to improve the quality of education for millions of students, nationally and globally (World Bank, 2021a; CONECTA, 2021a; UNESCO, 2021c; UN, 2022a; GPE, 2022; CONECTA, 2021b; OECD, 2021b; UNESCO, 2022f). For example, the Global Partnership for Education (GPE, 2022) which works with more than 76 partner countries has mobilized more than \$11 billion (USD) for education. This was part of its campaigns to raise funds needed to implement strong partnerships and strategies to increase global education and financing. The UN (UN, 2022d) has made a global call for investment in the future of education by supporting the Paris Declaration on Education within the 2030 Agenda for Sustainable Development. World leaders (which includes Heads of State and Government, and Ministers of Education) from more than 40 countries have also united

and adopted the Paris Declaration to include amongst the many initiatives, the goal to increase international aid to education and make it a priority to reach the target of 0.7% of donor GNP in terms of official development and assistance, and prioritizing education as the most powerful investment for our common future (UNESCO, 2021c, 2022e).

Instructional and infrastructural- wise, it is worth mentioning the educational partnerships and consortiums that have resulted and oscillated to the provision of blended pedagogies or instructional strategies (Okoye et al., 2021, 2022b), to digital skills training for the teachers especially in low-tech settings (Laufer et al., 2021), free training in digital literacy or computer skills for employment and entrepreneurial thinking, that all aim to connect the teachers and the students with the digital-savvy generation/education, new work reality, or yet post-pandemic education (Corvello, de Carolis, Verteramo, & Steiber, 2021), and preparing them for the most in-demand digital jobs, distance education, and life-long learning at large (UNESCO, 2021a).

2.4. Future of education: Ethical aspects and practice

Re-imagining better forms of using digital technologies in education or the future of education cannot be done without also imagining the ethical considerations to its best use and practices. According to the world's largest organization dedicated to advancing technology for the benefit of humans (IEEE) (IEEE Standard Association, 2022), "a key part of digital transformation comes in addressing the accompanying risks or de-risking". For educators to gain an adequate and effective digitized-education, all stakeholders (higher education institutions, teachers and students, educational community, policymakers) must take part or be part of the process of defining/designing the context, conditions, and purpose for which the resultant methods, technologies, or data and information are used (Facer, 2022; IEEE Standard Association, 2022; Slade and Prinsloo, 2013). Concerned with the question of what education may become or might be in the future, particularly the transformations enshrined and envisaged by the SDG4 (quality education goals), the new and emerging educational technologies or models must be designed and implemented to inflame and stimulate critical dialogs about its intentions or use for futures' practices (Facer, 2022). Ranging from issues that have to deal with their Reflexivity and Multiplicity, to Transparency, Curating or Unpicking harmful practices, Repairing, Re-balancing, and formation of strategies of Collective aspiration, to Intergeneration or co-construction between the different actors reimagining or acknowledging the fact that the future of digital technologies will be experienced differently by the different generations. There are also issues of What would be the emerging practices of care, learning, and responsibilities, Organizing and coordinating of the devices and platforms, Limiting pathological speculations, cultures or educational values, and Care for the distinctive laity and regenerative potential of education in general (Facer, 2022).

Didactically, a more heuristic and cross-sectional analysis, such as the one done in this current study, must be done to get a more conceptual outlook on the impact of the use of digital technologies in fostering the teaching and learning processes/outcomes. Apparently, those include a holistic approach to defining conditions under which the educational datasets are collected, stored, and analyzed. Indeed, not only does the so-called data-driven methods foreseen to offer context-appropriate solutions in the design of educational frameworks or models, but will also eventually cater to the sort-after new and better ways to track, monitor, and improve the educational systems or curriculum. Moreover, while it can be said that the algorithmic decision-making (data-driven) methods and models offer huge potential, especially when adequate procedures such as sensing, processing, acting, and learning are put in place (Miller, 2018). It is important to mention that the different actors must likewise acknowledge the risks and ethical concerns that are attached to such methods or use of the new and emerging learning technologies in education (Prinsloo, 2017; Okoye et al., 2020).

To summarize our review of the literature, *digital transformation* and *partnership* amongst the different stakeholders can play a vital role in creating synergies to rebuild the educational ecosystem, globally (GPE, 2022). The *new innovative* or *technological solutions* are perceived by the international and local communities, as one of the pertinent ways to achieving the enshrined “technological transformations” for a quality, equitable, and inclusive education (UNESCO, 2021b; ICFE, 2021). Many of educational organizations such as UN, in particular UNESCO, are working to promote the national capacities of the different institutions by actively promoting “education for sustainable development”, including other critical aspects of education such as the exchange of best practices, data collection, research and development (R&D), etc (UNESCO, 2022b; UN, 2022a). The increasing availability of open data and new technologies (such as AI, machine learning, big data, and cloud computing) are lowering the cost of using or adopting the learning technologies, thus, underlining the vast benefits and promising use and application of the digital technologies in education (UN, 2023). The key educational objectives must be to harness high-, low-, and even no-tech or no-cost solutions based on the individual learning demands or settings to foster the process of learning (Global Education Coalition, 2022). Moreover, *educational innovation* must not be limited to digital technologies or transformation alone, but should be encouraged throughout the education ecosystem (Global Education Coalition, 2022).

3. Methodology

The research method of this study is based on state-of-the-art education and assessment models for learning such as the Descriptive decision theory, Futures literacy (FLF), and 8-Affordances frameworks (Mulla et al., 2019; Chandler, 2017; Baucells and Katsikopoulos, 2011) that address both the need to using multidisciplinary components to

identify different challenges and opportunities to the use of digital technologies in education (digitized-education) and improvement of the teaching/learning outcomes.

3.1. Data sampling

The students’ opinion survey (ECO, 2024) used in this study is a SET instrument designed in a higher education setting to collect information about the teaching performance and learning outcomes of the students (see Appendix I). The instrument is an institutional survey administered online and maintained by the host institution for several years primarily for the purpose of teaching evaluation, and has also been validated in previous studies (Montemayor-Gallegos, 2002; Salinas and Martínez, 2018; Hernández, 2013). The items the study analyzed (see Experimental setup - Section 3.2, and Figs. 1 and 2) were completed by the undergraduate students during and after their respective courses offered across the 26 national campuses of the university in Mexico. As shown in the descriptive statistics (Figs. 1 and 2), we analyzed the data collected between the period of 2019 to 2021 based on the evaluation or feedback from the students concerning the new educational model called TEC21 model (TEC, <http://modelotec21.itesm.mx/files/folleto-modelotec21.pdf>; TEC, 2021, 2020a, 2023a). The TEC21 educational model (TEC, 2023a) applied by the host institution involves the design of study plans or curriculum that incorporate as a central unit of learning: the challenge-based learning (CBL) (Leijon, Gudmundsson, Staaf, & Christersson, 2022; TEC, 2023b; Torres-Barreto, Castro-Castaño, & Melgarejo, 2020) and the flexible digital learning (MFD) (TEC, 2021, 2020b) in which the students develop disciplinary and transversal skills by solving challenges linked to real-world problems and demonstrating their mastery through various evidence of learning.

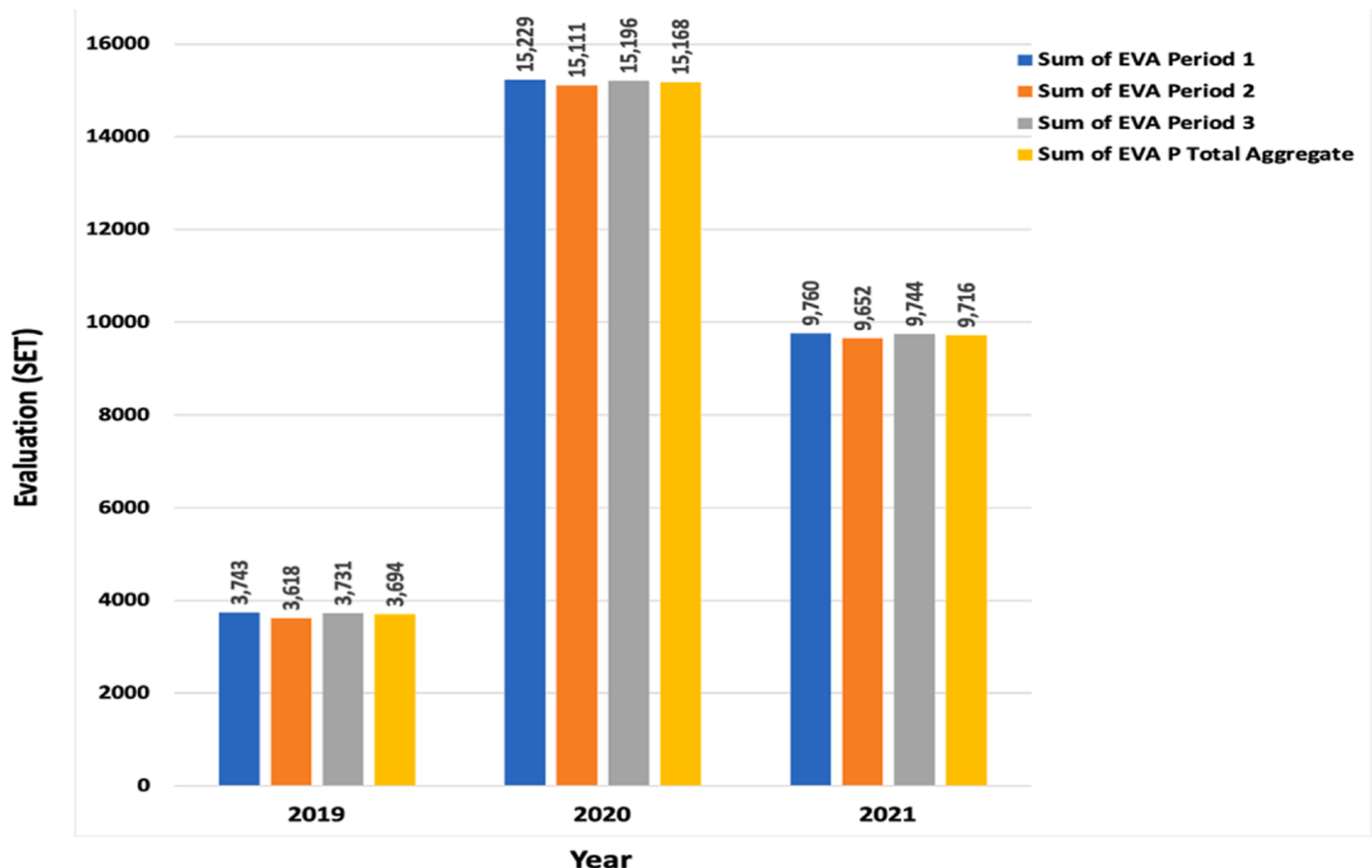


Fig. 1. Distribution of survey result (SET) broken down by sum of evaluation period and year.

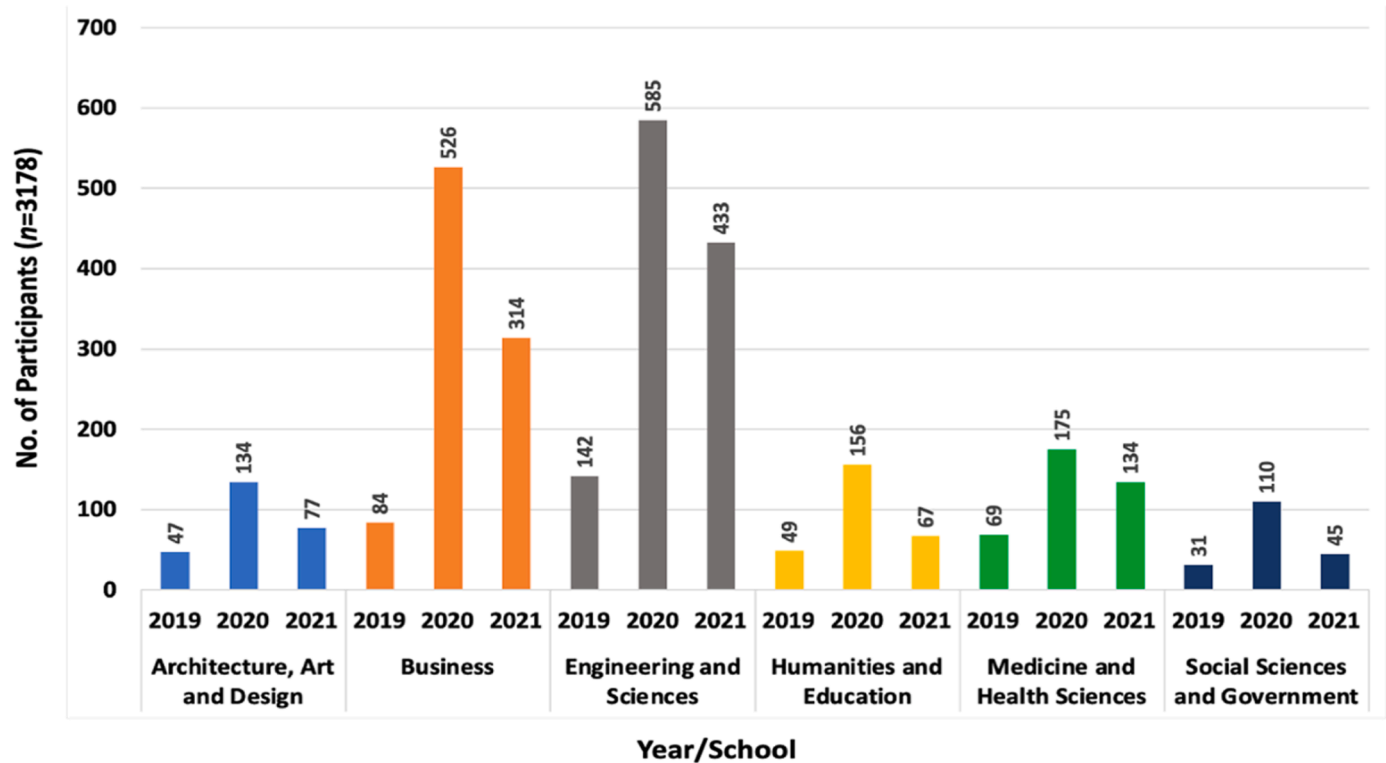


Fig. 2. Distribution of survey (SET) data broken down by School and Year.

To describe the research instrument and sample selection criteria (Figs. 1 and 2), the students were asked to complete the questionnaire in quartiles covering both during- and -end of their courses (i.e., Evaluation (EVA) Period 1 after 6 weeks, EVA Period 2 after 12 weeks, EVA Period 3 after 18 weeks, and a final evaluation EVA P Total Aggregate provided at the end of the courses). The analyzed data also include information about the six schools/disciplines enrolled by the students (Architecture Art and Design, Business, Engineering and Sciences, Humanities and Education, Medicine and Health Sciences, and Social Sciences and Government) (see Fig. 2). The sample selection criteria (inclusion and exclusion) were based on each participant must be registered on a program (school) and must have completed all the administered questionnaire (EVA periods) and reached end of course to be included in the study.

The study considered in each category of the analyzed items, a minimum number of 30 participants ($n \geq 30$) deemed scientifically acceptable and adequate to produce statistically significant results (Roscoe, 1975). For the individual items, the evaluation periods (EVA) were continuous values measured between 0 to 10, with 0 being the lowest and 10 being the highest. While the years and schools are represented as categorical variables, with the year considered as interval of the time or period (categorical) between 2019 to 2021.

The study analyzed a total sample of $n = 3178$ after preprocessing and cleaning the data by removing incomplete information (see Fig. 2). As shown in Fig. 1, the response rate for the participants across the years was approximately 14% in 2019, 53% in 2020, and 34% in 2021, respectively. Considering the schools (Fig. 2), we note that schools like Engineering and Sciences (1160 out of 3178) and Business schools (924 out of 3178) received the highest response rate in comparison to other schools, with the largest response recorded for the year 2020 in both cases (Engineering and Sciences: 585 out of 1160, and Business school: 526 out of 924) (see Fig. 2).

The reliability of the analyzed data and survey item was tested using Cronbach's Alpha test ($\alpha = 0.628$) which showed acceptable for conducting the data analysis (Taber, 2018). The parametric test

(MANCOVA) was applied considering the estimated minimum sample size we deemed scientifically acceptable ($n \geq 30$) (Roscoe, 1975) for conducting the research experiment and analysis, compared to the total sample size ($n = 3178$) the study used.

Ethically, the names and IDs of the participants were withheld for anonymity and to get an unbiased analysis of the data. The survey used a random sampling (probabilistic) method, thus, giving each participant an equal chance of being selected in the study. The students upon completing the survey were informed about the purpose for which the questionnaire is being completed and were not directly involved in the study's analysis.

3.2. Experimental setup

The authors present and describe the rationale for using the multivariate analysis of covariance (MANCOVA) and multiple pairwise comparisons (post-hoc) test/method to analyze the effect that the evaluation periods (2019–2021) and type of school (discipline) have on the SET evaluations or scores (EVA Period 1, EVA Period 2, EVA Period 3, EVA P Total Aggregate). MANCOVA (Li and Chen, 2019) is a type of analysis of variance (ANOVA) (Connelly, 2021), known as *N*-Way ANCOVA, where *N* represents the number of independent variables (e.g., Year and School/Discipline) being tested against the multiple dependent variables, e.g., EVA Period 1, EVA Period 2, EVA Period 3, and EVA P Total Aggregate (see Table 1).

In this vein, the statistical method and analysis, dataset variables, and conditions for the selected test (MANCOVA), and its appropriateness for answering the research question and data analysis (see Sections 1.1 and 4) are defined as follows, with a description of the method/test explained in Table 1.

- To determine the effect that the evaluation year (2019–2021) have on the SET scores (EVA Period 1, EVA Period 2, EVA Period 3, EVA P Total Aggregate), and how those may differ based on the test outcomes.

Table 1

Experimental setup and rationale for conducting the MANCOVA test.

Measurement	Variable	Test	Method/test description
Effect of Year and School/ Discipline on the students Evaluation scores or performance	• Evaluation year (2019-2021) = IV • Type of School/ Discipline = IV • SET scores (EVA 1, EVA 2, EVA 3, EVA Total Agg) = DV	Analysis of variance or test of mean effect by considering the Multiple set of Continuous Dependent variables (EVA 1, EVA 2, EVA 3, and EVA Total Agg.) analyzed against two Independent categorical factors and covariates (Year and School/ Discipline)	MANCOVA with multiple comparisons (post-hoc) test to determine whether the Year and School factors have an impact or are associated with the SET scores by the students, and does the impact, if any, differ by the independent groups of variables

Note: IV = Independent Variable, DV = Dependent Variable

- To determine the effect that the type of school or discipline has on the SET scores (EVA Period 1, EVA Period 2, EVA Period 3, EVA P Total Aggregate), and how those may differ based on the test outcomes.

Like the many other types of parametric tests or statistical procedures (Turner, Samaranayaka, & Cameron, 2020), the main assumptions the study took into consideration for conducting the MANCOVA test are summarized below and in Table 1.

- Independence of cases and homogeneity of variances: The multiple co-linearity test we conducted for the individual survey items (see Correlation matrix - Appendix II(A)) shows that the analyzed items are individually drawn from the data sample or population, with the ANOVA results for Year ($F=15.779$, $p=.000$) and School/Discipline ($F=2.276$, $p=0.059$) showing that the variance among the groups holds.
- Reliability or normality of data: Cronbach's Alpha ($\alpha = 0.628$) and minimum sample size ($n \geq 30$) (Roscoe, 1975) scientifically considered acceptable for conducting the parametric test (MANCOVA). Total sample size used equals to $n = 3178$.
- The independent variables (Year and School/Discipline) consist of more than two independent groups (i.e., 2019-2021, and 6 categories of schools) where: the independent variables are categorical, and the dependent variables (i.e., SET scores) are continuous scales, respectively.

4. Data analysis and results

The results of the method and data analysis defined in the experimental setup (Section 3.2) are presented in Tables 2, 3, and 4, and Figs. 3 and 4.

The results reported in Tables 2 and 3 show that there were significant differences in the students' evaluation scores across the years and schools ($p \leq 0.05$). The test of between-subjects effects (Table 3) shows

Table 2

Result of the MANCOVA analysis for year and school/discipline vs evaluation periods (EVA).

Factor	F	Error df	Pt. Eta Sq.	Sig.
Year	7.095	6340.00	0.009	.000*
School/Discipline	2.882 ^b	3169.00	0.004	.021*
Year * School/Discipline	4.023	6340.00	0.005	.000*

Significance level (*): $p \leq 0.05$, Dependent var. = EVA periods**Table 3**

Test of between-subjects effect for year and school/discipline vs evaluation periods (EVA).

Factor	Dependent Var.	Sum of Sq.	Mean Sq.	F	Pt Eta Sq.	Sig.
Year	EVA Period 1	5.81	2.90	4.33	0.003	.013*
	EVA Period 2	24.40	12.20	15.19	0.009	.000*
	EVA Period 3	1.83	0.91	1.44	0.001	.235
	EVA P Total Agg	4.45	2.22	4.65	0.003	.010*
School/ Discipline	EVA Period 1	3.98	3.98	5.93	0.002	.015*
	EVA Period 2	0.05	0.05	0.07	0.000	.789
	EVA Period 3	0.11	0.11	0.18	0.000	.665
	EVA P Total Agg	0.77	0.77	1.61	0.001	.204
Year * School/ Discipline	EVA Period 1	5.80	2.90	4.33	0.003	.013*
	EVA Period 2	3.78	1.89	2.35	0.001	.095
	EVA Period 3	0.93	0.46	0.73	0.000	.480
	EVA P Total Agg	0.15	0.07	0.15	0.000	.854

Significance level (*): $p \leq 0.05$ **Table 4**

Multiple pairwise comparison of the between-subjects effect considering the evaluation periods broken down by Year.

Item	Year (A)	Year (B)	Mean Diff (A-B)	Std. Error	Sig.
EVA Period 1	2019	2020	-.149	.045	.003*
		2021	-.236	.047	.000*
	2020	2019	.149	.045	.003*
		2021	-.087	.032	.020*
	2021	2019	.236	.047	.000*
		2020	.087	.032	.020*
EVA Period 2	2019	2020	-.398	.049	.000*
		2021	-.456	.052	.000*
	2020	2019	.398	.049	.000*
		2021	-.058	.035	.303
	2021	2019	.456	.052	.000*
		2020	.058	.035	.303
EVA Period 3	2019	2020	-.166	.044	.000*
		2021	-.259	.046	.000*
	2020	2019	.166	.044	.000*
		2021	-.093	.031	.009*
	2021	2019	.259	.046	.000*
		2020	.093	.031	.009*
EVA P Total Agg.	2019	2020	-.240	.038	.000*
		2021	-.323	.040	.000*
	2020	2019	.240	.038	.000*
		2021	-.084	.027	.006*
	2021	2019	.323	.040	.000*
		2020	.084	.027	.006*

Significance level (*): $p \leq 0.05$, p-values adjusted = Bonferroni method

the variations in the evaluation (EVA) scores, whereby the majority of the differences were found for the year construct ($p \leq 0.05$), except for EVA Period 3 that presented insignificant ($F=1.44$, $p=.235$). Likewise, when considering the schools/discipline, we found that the type of school or course the students' study does not significantly impact the evaluations provided by the students ($p \geq 0.05$) except for EVA Period 1 that was significant ($F=5.93$, $p=.015$) (Table 3). Also noteworthy is the fact that the marginal means of effect by considering the combination of the Year*School/Discipline (Table 3) as co-variates showed that the EVA Period 1 varied in the SET scores ($F=4.33$, $p=.013$), as opposed to EVA Period 2, EVA Period 3, and EVA P Total that presented non-significant ($p \geq 0.05$) (Table 3).

Table 4 and Figs. 3 and 4 represent the results of the multiple pairwise comparisons (post-hoc) test we performed to determine where the significant differences we have found may lie in the data.

As gathered in Table 4, we note that the significant differences found for the EVA scores (see Table 3) was across the analyzed years (i.e., between 2019 to 2021) ($p \leq 0.05$) except for the period of 2020 vs 2021

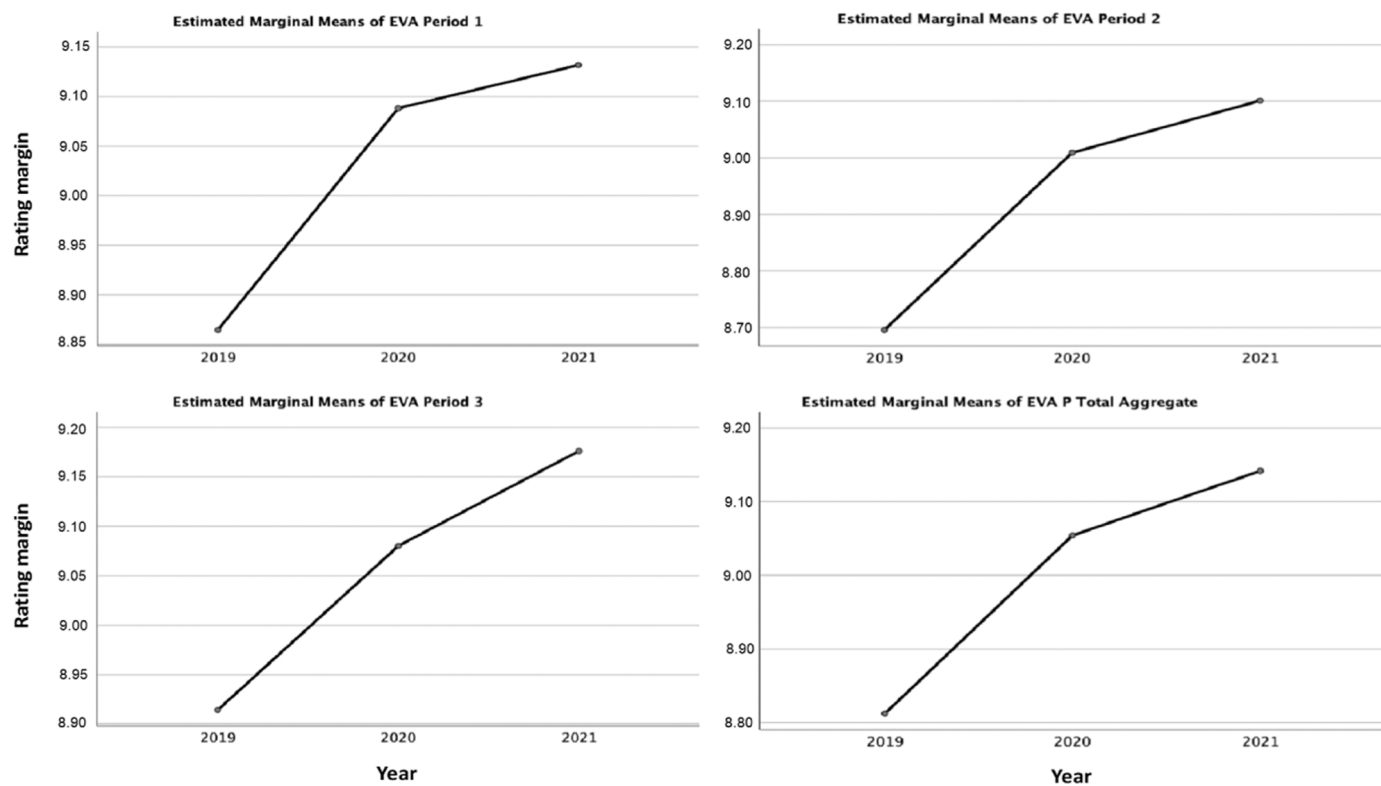


Fig. 3. Marginal mean of effect for the Evaluation periods (EVA) broken down by Year (2019–2021).

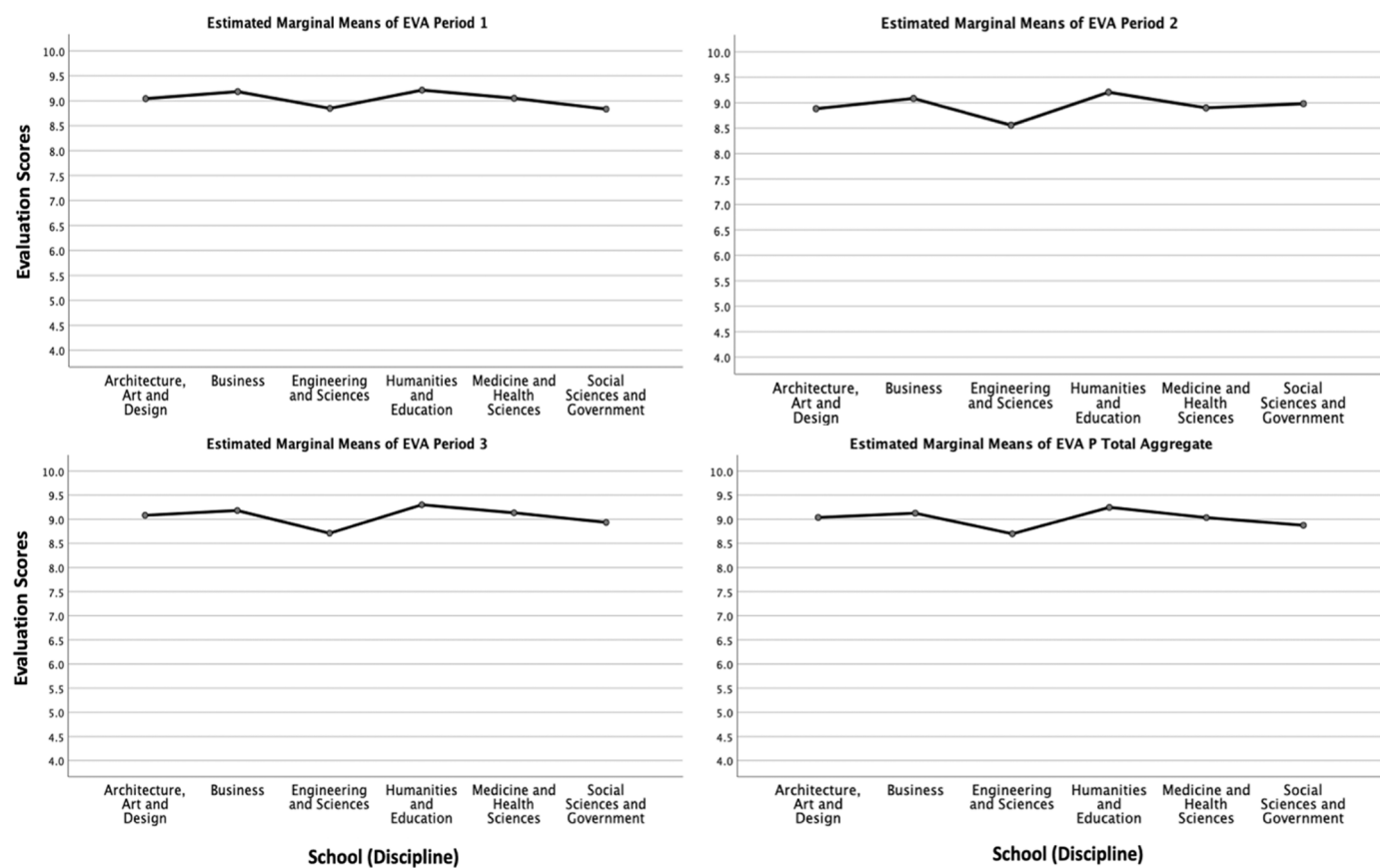


Fig. 4. Marginal means of effect for Evaluation periods (EVA) broken down by School/Discipline.

for EVA Period 2 ($p = .303$) therein we found that there were no differences in the way the students have evaluated their learning performance or outcome (see Table 3).

Figs. 3 and 4 represent the estimated marginal means of effect for the years and schools/discipline broken down by the Evaluation periods (EVA). The figures (Figs. 3 and 4) show the thresholds or margins in the way the students have rated their learning outcome over the years (2019 to 2021) and across the schools/disciplines, which are empirically discussed in detail in the Discussion (Section 5).

5. Discussion

To discuss the study findings and results (Tables 2, 3, 4 and Figs. 3 and 4), we note in Fig. 3, that there was an exponential growth or threshold in the students' satisfaction across the years (2019–2021) with the least margin ($\sim 8.0 \pm$) observed for 2019, and max ($\sim 9.0 \pm$) observed for 2021 periods, respectively. While it can be said that the year factor has a significant effect on the student's evaluation ($p \leq 0.05$), and the type of school or discipline does not systematically influence (impact) the evaluations provided by the students ($p \geq 0.05$) (see Tables 2 and 3). On the other hand, when considering the fact that the students have likewise expressed high satisfactions across all the schools (Fig. 4), it is noteworthy to mention the largest satisfaction ($\sim 9.0 \pm$) was observed for the Humanities and Education, and Business schools, and the least ($\sim 8.0 \pm$) for the Engineering and Sciences, and Social Sciences and Government schools, respectively. The exponential growth in the students' satisfaction or evaluation we observed for the analyzed periods or years (2019–2021) (Fig. 3) and schools (Fig. 4) with higher margins of above $8.0 \pm$ (see Figs. 3 and 4), may be linked to how appreciative and prophylactic the students had eventually become following the use, acceptance, and wide adoption of the new and emerging learning technologies (digitalized-education) under the new learning circumstances (OECD, 2022a; UNESCO, 2022a; UIS, 2023).

Didactically, a number of factors may explain or contribute to these findings. First, the new modality of learning (virtual or digital) introduced to the students particularly during the recent global pandemic (2020–2021) may typically have affected the level of access and support given to the students whose courses/learning involved laboratory work, as opposed to the theoretical groups who are able to get full access and support online (continuous teaching/learning). This can explain the reason why the study found the least, yet positive evaluations ($\sim 8.0 \pm$) for schools like Engineering and Sciences in comparison to the higher rating ($\sim 9.0 \pm$) observed for the Humanities and Education and Business schools. Likewise, the differences in the evaluations scores or ratings can be evidenced in the multiple co-linearity test we conducted for the survey items (see Appendix II (A) and (B)), therein we found that the significant relationships ($p \leq 0.05$) between the SET scores and the analyzed factors was for the school/discipline (see Appendix II (B)). It is also noteworthy to mention the large number of questionnaires completed by some schools across the years (Fig. 2) in comparison to others, which might also represent to have a significant impact on the SET outcomes. Finally, the several challenges and changes the students faced due to the shift from face-to-face to remote classroom (e.g., conducting practical engineering and science labs, digital representation of mathematical equations or formulas, etc.), may have inadvertently not allowed the students in those types of courses requiring face-to-face interactions to effectively participate in the teaching/learning processes during those years that appeared significant (Tables 2 and 3), e.g., amidst the recent global pandemic between 2020–2021, retrospectively.

5.1. Implications of this study

The need for “digitalized-education” has never been more and more, made clearer, alongside the several innovations that have spanned across the years. Considering the socio-technical challenges, instructional awareness, and perceived usefulness of the digitalized-education,

the authors note that it is noteworthy to shed light on the several implications and promises of drawing on the new (educational) technologies and data in the delivery of education or learning assessment. Educators should be attentive of how the new models of learning run the risk of exacerbating inequalities or limited access to learning, especially for those courses that require a significant level of face-to-face interactions or lack/limited access to the digital platforms or networks highlighted both in our review of the literature (see Section 2) and discussion of the results of this study (Section 5).

To emphasize on the current studies' contribution to understanding both the theoretical and practical implications and application of the “digitized-education” and “educational-generated data”, including the effectiveness and strength of the research methodology and analysis (see Section 3); we provide a clear discussion between the results of the study and the educational/practical implications. In triangulation with this research study and outcome (see Sections 1.1 and 4) and the recent dialog or peer learning for the international community of practice among the educators and statistics report by UNESCO (UIS, 2023), which looks at data for education or guide for policymakers by leveraging educational data; we note that the UNESCO's Institute for Statistics (UIS) (UIS, 2023) has pointed out that learning assessments, such as the one done in this present study, offers valuable insights into the quality and equity of education by harnessing or uncovering important aspects and didactical questions (see Sections 1.1, 2, and 5.1.1–5.1.3) that can be addressed using data from learning assessments, such as the one done in this study. Idiosyncratically, the UIS (UIS, 2023) report looks into why the congruence of the “learning assessment or evaluation” and “leveraging education data”, which forms the main components and methodology of this study, are relevant for education policy planning and SDG4 (that promotes quality of education and monitoring) (UIS, 2023).

Along these lines, some critical aspects of the digitized-education and the effective use of educational-generated data, its anticipated transformations, and implications for the future of education are explicitly discussed in the next sections (Sections 5.1.1–5.1.3) of this paper in connection to the results and study outcome.

5.1.1. Implications of digitized-education for global schooling and life-long learning

It is hoped that digitalized-education and the associated practices and impact uncovered in this study will help to inform and heighten the use of educational-generated datasets and technologies toward achieving the quality of education (World Bank, 2021a; Mulla et al., 2019; UIS, 2023; UNESCO, 2021c). Dated back in 1967, UNESCO produced one of the first analytical reports on education worldwide (Draxler and Alexandra, 2014; Coombs, 1968; Cordasco, 2008), therein it announced a worldwide crisis in education, and attributed it to factors such as the tension between “learning demand” and “availability of learning resources”, to societal inertia, and not enough or limited pedagogical understanding or information about education as a system (Draxler and Alexandra, 2014). One of the implications of this current study is that it shows that the educational-generated datasets can be used to understand and draw insights that can be used to inform and improve the educational processes and decision-making strategies (see Sections 3 and 4). In other words, the aforementioned challenges which idiosyncratically can be supported through effective use and analysis of the educational data, such as the one done in this study, has proved effective and affirmed by the existing studies in recent time (UIS, 2023; Raffaelli, Manca, Stewart, Prinsloo, & Sangrà, 2020; Cech, Spaulding, & Cazier, 2018).

Fast track to the 21st-century education, it can be said that the “digital technologies” have shown its prominence in facilitating the long-desired shift from “learning as passive content acquisition” to “learning as active knowledge co-creation and communities of practice” (Mulla et al., 2019). Educational organizations like UNESCO (Mulla et al., 2019; UNESCO, 2021c) has continued to point out there is a real

opportunity to change the pattern or way digital technologies are used for learning, e.g., through the promotion of collaborative peer learning, and learning analytics for student assessment, etc (Mulla et al., 2019). The tendencies, on the one hand, means that educators must also do this right, by drawing from the most relevant and credible information and sources in the field, e.g. by leveraging the educational datasets, and essentially must align the developed methods and models with international guidelines, practices, or global point of view (OECD, 2021a; World Bank, 2021a; Mulla et al., 2019; OECD, 2022b; UNESCO, 2021b, 2021c, 2020a, b). As an example, the study by Jones (2022) noted that educators are now tapping into the so-called “digital revolution” and are adopting new technologies to help learners reach their full potential, and studies such as this present one will utmostly serve as both a methodological/practical road map and incentive to actionable/informative studies in the topic area.

It is likewise argued that lately, the education systems are due for an update, globally (Jones, 2022). Therefore, achieving “digitized-education” which forms the main topic in this study, and consequentially, *quality of education* as the overlapping by-product, is currently not a national or regional problem, but an international and investment challenge (UN, 2022a; Watkins, 2022). Collective partnership or joint effort amongst the involved stakeholders or institutions, has never been more and more ever than before, emphasized as one of the most promising ways to move forward (OECD, 2022b; UNESCO, 2021b; Okoye et al., 2022a; UNESCO, 2021c, 2022e; GPE, 2022). To note, in some countries, collaboration with universities and other partners (research centers, firms, government parastatals, etc.) has been vital, and has offered space for bottom-up initiatives to foster and ensure effective learning or education (Schneegans, Lewis, & Straza, 2022). Noteworthy is the fact that UNESCO in its Science Report has alarmed that the clock is ticking for transitioning to “digital societies” (Schneegans et al., 2022). Recent study by Wang and Cheng (Wang and Cheng, 2021) has also noted that the advancement or acceleration of the “digitized-education” will probably require complementary investment in data, skills, and digitalized workflow, among other educational innovations.

5.1.2. Outlook on pedagogical or instructional implications of the digitized-education

To consolidate the process of digital transformation of the higher education institutions, technological design and upskilling programs (for example, programs such as TechSavvy) (Techvolution, 2022), must be pioneered to adapt to the digital age, and aim to strengthen/improve the teachers and students’ skills by acquisition of new skills didactically oriented toward achieving the most benefits from the education ecosystem. Educational datasets can be leveraged to enable the informative and pedagogical values of such systems (UIS, 2023). The future learning settings or “classrooms of the future” per se are presumably projected around “self-paced” and “personalized learning”, and emerging technologies and tools such as blended learning, flipped classrooms, classroom 2.0, artificial intelligence, chatbots, and video conferencing/learning has been foreseen as a central backbone or auxiliary steps in these directions (Jones, 2022; Wang and Zhu, 2019; Xu, Yuan, & Liu, 2021; Godwin-Jones, 2020; IEEE, 2020). In turn, these tools and technologies can help create an “open and sustainable architecture” or framework for learning both from the local and global perspective (Patrinos and Bustillo, 2022). Moreover, the educational systems update or yet the “digitalized-education revolution” is perceived to eventually mean (i) greater reach to the different audiences and stakeholders, e.g., educators, teachers, students, education society, etc., (ii) AI methods/technologies revolution, (iii) global citizenship or life-long learning for the learners throughout life, and (iv) diverse careers/skills for work and entrepreneurship (Jones, 2022).

An estimated 98% of educators has perceived the new and emerging (educational) technologies as an essential component to enabling personalized learning experiences for the students (Jones, 2022). The

concomitant TEL-based models have presented not only to support effective learning and increase the performance for the students (Khojasteh et al., 2022; Moorhouse and Wong, 2022), as seen in the results of this study (see: Section 4). But likewise, the learning technologies have been reported to increase the students’ satisfaction by 91%, and achievements by 82%, which validates not only the results of this study, but also the reason why educators are increasingly using AI and the new technologies to foster the teaching/learning processes (Jones, 2022).

However, at the same time, it is noteworthy to mention despite the many innovative (educational) technologies that have bloomed over the years, especially during the recent Covid-19 pandemic; the clear lessons learned by educators and all the concerned stakeholders remained that the educational systems that did not pay adequate or sufficient attention to the use of the “educational-generated datasets” or “educational technologies” (TEL-based education), learned the hard way that it needs to be included as part of the schools’ ecosystem or instructional/decision-making strategies (Lennox et al., 2021; Patrinos and Bustillo, 2022; OECD, 2022d).

5.1.3. Economic and infrastructural development implications and social-technical standpoints

Students affected by the schools’ closure due to the recent global pandemic, will attain 3% less income during their professional lives (UNDRR, 2022). This comprises an approximated loss of 1.5% of GDP over the remainder of the present century (UNDRR, 2022). The latest evidence from the United Nations Office for Disaster Risk Reduction (UNDRR) (UNDRR, 2022) has accentuated “infrastructural investments in schools” as one of the most pertinent ways to provide the required economic development and returns for society. This includes the much-needed economic recovery (post-pandemic), achievement of the sustainable development (SDG4), and strengthening or modernizing of the technology/infrastructural transformation for instructional purposes revealed in our review of the literature (OECD, 2021a; World Bank, 2021a; Lennox et al., 2021; OECD, 2022b; UNESCO, 2022e; OECD, 2021b; Patrinos and Bustillo, 2022; Global Goals, 2024). In a similar case, UNICEF (UNICEF, 2022) upon recognizing the technical/socio-economic challenges facing educators, policymakers, and instructional planners, particularly in resource-constrained countries, acknowledged the fact that despite efforts to mitigate the “digital divide” through the use of multiple modalities (e.g., hybrid modes) and a mix of high-, low- and no-tech options, that a significant number of the students were still left out, therefore, deepening the already significant pre-existing inequalities (Lennox et al., 2021; O’Hagan, 2022). In those recommendations, Lennox et al (Lennox et al., 2021). note that the didactical tools (educational technologies) must be paired with “investments in the people” expected to use them, including a strengthened educational-data literacy or ecosystem, as highlighted in the results of this study and our review of the literature (LALA, 2020; Raffaghelli et al., 2020).

Furthermore, while science and education are inadvertently paying the price of the economic downturn, many governments, financial investors, and policymakers are also prioritizing higher education (Facer and Selwyn, 2021; World Bank, 2021a; OECD, 2022b; UNESCO, 2021b, c, 2022f; Schneegans et al., 2022). In the wider spectrum of scientific research, it can be said that this study has brought to light the attention of educators and organizations, especially in line with the United Nations’ Foundation’s Digital Impact Alliance (UN, 2022e) objectives that recognize nine principles for “digital development”, such as the digitalized-education, that is capable of enabling ICT-led transformations, inclusion, and sustainability in education, to include: (i) designing with the user (ii) understanding of the existing ecosystem (iii) designing for scale (iv) building for sustainability (v) be data-driven (vi) use of open standards, open data, open-source, and open innovation (vii) reuse and improve (viii) address privacy and security, and (ix) be collaborative. Indeed, our review of the literature and results not only emphasizes the need for studying the effects of the new dynamics and/or

opportunities offered by the “digital-for-development” education or instructional strategies, but also demonstrates the benefit of harnessing the educational datasets (e.g., SET) to inform and drive the educational process forward (Mulla et al., 2019; Okoye et al., 2022a; UIS, 2023; Laufer et al., 2021; Heeks, 2020).

Socio-technical-wise, the methodology and results of this study prove that the educational datasets (e.g., SET) can be extracted, analyzed, and transliterated into valuable information and actionable plans that can be used to support the educational strategies and decision-making by the institutions. Educators and different educational bodies are stepping up to the plate in terms of nurturing the digitized-education, partnerships, and infrastructural development (GPE, 2022; Schneegans et al., 2022). However, there are still global conundrums or issues tied to the notion of “without investment, no innovation” that has to do with different stages of infrastructure or learning preparedness, learning sustainability and scalability, and/or gradual convergence of the digital knowledge ecosystems (Schneegans et al., 2022). Therefore, the authors believe that the outcome of this present study serves as a methodological incentive to further research that deals with *digital technology transformation* and its implications for the future of education. Technically, *innovating education* through the lens of transforming beyond the pandemic (post-pandemic education) means four key pillars: (i) enabling learners to learn anywhere, anytime, and at any place (ii) building adequate skills, particularly digital skills, for the future of work (iii) supporting educators, teachers, and students to thrive in the current digital-savvy generation, and (iv) building back equality, both in terms of the digital divide, and socio-technical, demographic and gender dispositions (UNESCO, 2021b; O’Hagan, 2022; Techvolution, 2022). It is hoped that educators can offer and encourage social-economic and entrepreneurship development by transforming the global communities through digitally-enhanced (TEL) education (Frezzo, 2022; O’Halloran, Opp, & Rastogi, 2022; Yoon, 2022).

To summarize the discussions and implications of this study. The use of educational technologies for learning and leveraging of educational-generated data has proved effective and indispensable for the future of education and its transformation (UIS, 2023). However, the stakeholders (educators, governments, policymakers, financial investors, EdTech companies) must also pay a great amount of attention to the design and reimagining of the potential barriers and bottlenecks to its use and implementation (Okoye et al., 2022b) highlighted in this study (see Sections 2 and 5). For example, existing research has suggested actions for educators, different governments, and researchers to add (in coalition) with the edtech or AI initiatives; technological proposals and advance policies or documentations that can lead to the creation of an ethical observatory of AI for education (UNESCO, 2022c, 2021b, 2021b, 2021c; ICFE, 2021; Flores-Vivar and García-Peñalvo, 2023; Selwyn, 2022). Noteworthy, Flores-Vivar and García-Peñalvo (Flores-Vivar and García-Peñalvo, 2023) have mentioned disparities that may exist in the practical sense, by emphasizing uncertainties caused by the fear of some stakeholders who think, for instance, that AI or robots could replace human teachers, despite the fact that the impact of AI methods or digitalized practices in education has been relevant and pertinent in addressing to name but a few; the global sustainable development goals (particularly SDG4 that promotes the quality of education), the UNESCO’s 2030 Agenda, impact that the new and emerging algorithms and robotization are causing in the field of education, numerous benefits and opportunities it has presented to teachers and students, or what we now call the post-pandemic education era or digital-savvy generation (OECD, 2022a, 2021a; Facer, 2022; World Bank, 2021a; UNESCO, 2022b; Laufer et al., 2021; Okoye et al., 2022a; UN, 2022a; Flores-Vivar and García-Peñalvo, 2023; Selwyn, 2022). Interestingly, in response to the identified short falls in advance documentations and directives, the UNESCO has recently launched under its Global AI Ethics and Governance Observatory (UNESCO, 2023a) the Ethical Impact Assessment (EAI), a tool for recommendation on the ethics of artificial intelligence, which considers the entire process of designing, developing and deploying AI

systems by allowing for assessment of the risks before and after the system is released to the public or educational community at large.

At the interdependency level, education is one of the proven means by which information and understanding of the rapid advances in AI tools and methods (e.g., virtual reality, augmented reality) and other advanced technologies (robotics, cloud computing, Internet of Things) are being used, disseminated, and are transforming schools, economies, and industries, alongside with it challenging theoretical ideas about what it means to the human race (Flores-Vivar and García-Peñalvo, 2023). We must highlight the speculations that AI tools and methods must be used to augment and amplify human capabilities, and drive the (digitized) education ecosystem forward, but not replace them (Laufer et al., 2021; O’Hagan, 2022; Flores-Vivar and García-Peñalvo, 2023; Angel-Urdinola, Castillo, & Hoyos, 2022). Enshrined by UNESCO (UNESCO, 2023a, b), the potentiality or literacy of digital technologies or AI, particularly for quality education and delivery, requires both an awareness of its values and more knowledge about the potential of the digital technology (Chen, Chen, & Lin, 2020), as seen in this study and our review of the literature. The digitized-education or application of AI in education is believed to ultimately promote improved access to learning and its reach through the elimination of the different barriers and bottlenecks to learning (Laufer et al., 2021; O’Hagan, 2022; Okoye et al., 2022b), automation of administration and management systems and functions of the different institutions (Okoye, 2022; Mourad, 2017), optimization of the instructional methods or models for learning (Chiu, 2020), and increase in empirical/evidence-based research, decision-making, policies, and strategic planning and initiatives in education (Miao et al., 2023; UNESCO, 2023a,b; Chen et al., 2020).

5.2. Limitations and directions for future research

While the authors believe that the methodology and discussions in this study are suitable for conceptualized outlook on critical factors that impacts the use of the digital technologies in education (digitized-education) and the future of education, we acknowledge that there may also exist some limitations to the study. For example, whilst there are theoretical and socio-technical need for ground-breaking methods or suitable studies that harvest educational-generated data and information to improve the pedagogical practices, as done in this study; the authors note, on the other hand, that such (educational) transformations can come with the portentous task of ensuring that the users of the said resultant (educational) technologies and models are imbued with the necessary (digital) competencies, skills, and training, to help maximize the impact/adoption of the digitalized-education, which can form the basis for further studies.

Also, whilst the study has adopted the data-structure approach based on the descriptive decision theory, futures literacy, and 8-affordances frameworks (see Section 3) to study the impact and implication of the digitized-education particularly toward the future of education, there could be potentially many other ways to approach this topic which may have not been considered in this study. For instance, it may be worthwhile to mention that before addressing the different challenges associated with the “digital divide” or other technical/social factors limiting the performance/outcome of the digitalized-education especially for the teachers and students, globally. On the other hand, we note that adequate and alternative solutions or methods for unbiased replacing of the in-person encounters or practical interactions that hypothetically constitute a sizeable part of the present educational curriculum and models should be tackled to ensure an equitable and sustainable education in all senses.

Finally, we note that the comparative analysis of the students’ evaluation of teaching dataset done in this study are based on the SET instrument applied within the higher education setting and studied periods. Therefore, further studies or data may be collected and analyzed in other contexts or timeframes to complement the findings and results already provided in this study. For example, factors such as students’

bias in evaluation and/or type of program (e.g., undergraduate, graduate, general education courses, or professional courses) that might lead to different response rate or course evaluations may be another limitation that can be studied. While existing studies have noted that many higher institutions of learning rely heavily on the outcomes of the students evaluation of teaching (SET), e.g., as a decisive factor in promotion, long-term contracts, merits, and award-related decisions, and contract renewals for teachers, used in improving the different learning activities and decision-making strategies about the teaching and learning process, and determining the students' academic performance and assessment of teachers (Bälter, 2021; Talandron-Felipe, 2020; UNESCO, 2020a; OECD, 2021a). On the other hand, research has increasingly scrutinized or examined the impact and validity of the SET as an indicator of the teachers' and students' performance or assessment. For example, contemporary studies have provided new evidence on gender bias in teaching evaluations by noting several assumptions about differences in teaching qualities and performances of male vs female teachers, often presumed to differ in terms of teaching styles, capacities, and effects considering the students' perspectives and teaching as a career (Talandron-Felipe, 2020; Facer and Selwyn, 2021; Sönderlund et al., 2019). However, it is important to mention there is the need to explore the potential of the information from SET datasets in providing the ability to make effective educational-process-related decisions (e.g., monitoring of the teaching-learning process, and content recommendation, etc) (UNESCO, 2022a). In terms of the types of programs enrolled by the students which may lead to different response rates or course evaluation; we note that previous studies have shown that the students' category (e.g., undergraduates, postgraduates) can influence the extent or motivation levels by the students (Okoye et al., 2021). For instance, postgraduate students are exposed to advanced and more rigorous learning settings, and besides, could be more adaptable to different or adverse learning circumstances, which may not necessarily influence their rating of the teaching performance and outcome in comparison to the undergraduates, which may form another direction for further research.

6. Conclusion

This study applied the MANCOVA and multiple comparisons (post-hoc) tests to study the impact of digital technologies in education and its implication for the future of education and sustainable practice. It used SET data ($n = 3178$) collected in a higher education setting to investigate the main factors that influence the student's performance under the new learning modalities and circumstances. The study was based on data-structure approach it grounded on state-of-the-art education and assessment models or frameworks such as the descriptive decision theory, futures literacy (FLF), and 8-affordances (8-A) framework. The results of the study shows that there has been exponential increase in the students' satisfaction in use of the digital technologies for learning over the years. However, some schools or disciplines such as Engineering and

Sciences, and Social Sciences and Government showed to have experienced the most challenges, e.g., due to the new modalities or shift from face-to-face to remote/hybrid learning modes and settings that may not fully support the practical laboratory works. The study critically sheds light on the results and key findings by considering its main implications for the future of education and sustainable/scalable educational practice, particularly the SDG4 that promotes the quality of education. Consequentially, the outcome of this study can help educators, policy-makers, and governmental organizations to create an open and sustainable architecture or framework for learning both from the local and global perspectives or point of view. Future studies can adopt or extend the method defined in this paper to include other components or analyses that may have not already been done in this study.

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CRediT authorship contribution statement

José Francisco Enríquez de la O: Resources, Software, Validation, Investigation, Methodology, Project administration. **Jose Escamilla:** Resources, Supervision. **Sandra Dennis Núñez Daruich:** Funding acquisition, Investigation, Methodology, Resources, Software. **Raquel Castaño:** Project administration, Resources, Validation. **Samira Hosseini:** Resources, Supervision, Writing – original draft, Writing – review & editing, Project administration. **Kingsley Okoye:** Conceptualization, Data curation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing interests or personal relationships that could have appeared to influence the work reported in this paper.

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Originality and submission status

The authors confirm that this work is original and has not been published elsewhere, nor is it currently under consideration for publication elsewhere.

Appendix I. ECOA SET instrument: Students' evaluation under the flexible digital learning (TEC21 model)

Item No.	Factor	Question	Scale of measurement
1	Mastery of the subject matter - (DOM)	My teacher(s) showed mastery and expertise in the subject matter	1 to 10, Totally disagree = (0), Totally agree = (10)
2	Intellectual Challenge - (RET)	My teachers(s) challenged me to do my best (develop new skills, new concepts and ideas, think differently, etc.)	1 to 10, Totally disagree = (0), Totally agree = (10)
3	Adviser - (ASE)	The teachers(s) fostered an environment of trust and respect (i.e., regarding the interaction with my teacher and the advice received during the learning process)	1 to 10, Totally disagree = (0), Totally agree = (10)
4	Methodology and learning activities - (MET)	The support I received from my teachers(s) was adequate (answering questions, advice, feedback, etc.)	1 to 10, Totally disagree = (0), Totally agree = (10)

(continued on next page)

(continued)

Item No.	Factor	Question	Scale of measurement
5	Recommendation of professor - (REC)	In general, my learning experience with the teacher(s) was:	1 to 10, Deficient = (0), Excellent = (10)
6	Comment - (CMT)	What would you comment to a student who would like to enroll in the class with the teacher?	Open-ended
7.1	Learning guide (A): motivation and commitment to my learning, development, and overall growth - (APD)	The topics, activities and situations during the class: (A) Allowed me to learn and develop myself.	1 to 10, Totally disagree = (0), Totally agree = (10)
7.2	Learning guide (B): understanding of concepts in terms of their practical application - (PLV)	The topics, activities and situations during the class: (B) Are applicable and valuable.	1 to 10, Totally disagree = (0), Totally agree = (10)

Appendix II. (A). Correlation matrix (Multiple linear regression) table for the year and school/discipline factors vs SET evaluation scores

Co-linearity for Year and School/Discipline vs Evaluation scores (Method = Pearson Correlation)						
Dataset		Year	EVA Period 1	EVA Period 2	EVA Period 3	EVA P Total Agg
Year (n = 3178)	Year	1.000	.092	.131	.100	.135
	EVA Period 1	.092	1.000	.499	.458	.739
	EVA Period 2	.131	.499	1.000	.565	.821
	EVA Period 3	.100	.458	.565	1.000	.833
	EVA P Total Agg.	.135	.739	.821	.833	1.000
School/Discipline (n = 3178)	School / Discipline		EVA Period 1	EVA Period 2	EVA Period 3	EVA P Total Agg.
	School/Discipline	1.000	-.028	-.022	-.003	-.029
	EVA Period 1	-.028	1.000	.499	.458	.739
	EVA Period 2	-.022	.499	1.000	.565	.821
	EVA Period 3	-.003	.458	.565	1.000	.833
	EVA P Total Agg.	-.029	.739	.821	.833	1.000

Note: Sig. (1-tailed) = .000

Appendix II. (B). Correlation coefficient (ρ) results for year and discipline/school

Model (ρ)		Unstandardized Beta (β)	Std. Error	Standardized Beta (β)	t	Sig.
Year (n = 3178)	EVA Period 1	-.011	.027	-.013	-.398	.690
	EVA Period 2	.035	.028	.048	1.232	.218
	EVA Period 3	-.022	.035	-.027	-.642	.521
	EVA P Total Agg	.120	.075	.128	1.596	.111
School/Discipline (n = 3178)	EVA Period 1	.056	.054	.035	1.031	.303
	EVA Period 2	.079	.057	.055	1.398	.162
	EVA Period 3	.174	.070	.107	2.485	.013
	EVA P Total Agg	-.352	.151	-.189	-2.337	.020

Note: Significant levels (p ≤ .05)

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